Morphosyntax-Only Parsing: Czech vs. English

1 Introduction

This project asks how much dependency structure can be recovered from morphosyntactic information alone, and whether the answer differs between Czech and English. Dependency parsing is a good test case because the target structure is not directly visible on the surface. A parser must decide which word heads each other word and which dependency label links them. In many current systems, lexical identity, contextual encoding, and morphological information are combined, so it is hard to see how much of the final performance comes from each source. Here I isolate that question by comparing three versions of the same parser: lexicalized, morphosyntax-only, and mixed.

The comparison is interesting linguistically as well as technically. Czech is morphologically richer than English, with more overt inflectional marking and agreement. English is more analytic and often relies more on word order and lexical information. If morphosyntax-only parsing works much better in Czech than in English, that would suggest that overt morphology itself carries strong clues about syntactic structure. If both languages behave similarly, that would suggest that lexical identity is still doing most of the work even in a rich-morphology setting.

The parser is graph-based, scores head-dependent arcs explicitly, and uses maximum spanning tree (MST) decoding to enforce a legal tree. The evaluation is controlled rather than focused only on benchmark performance. The goal is to learn something about what morphosyntactic information contributes and how that contribution differs across languages. The hypothesis is that morphosyntax-only input should outperform lexicalized input in both languages, but that the gain should be much larger in Czech.

2 Data and Experimental Setup

The experiments use two Universal Dependencies treebanks: UD Czech-PDT and UD English-EWT. These datasets are appropriate because they provide dependency trees together with universal part-of-speech tags and morphological features in CoNLL-U format. That makes them a natural fit for graph-based dependency parsing and morphosyntactic representations.

The datasets used in this project are publicly available from the Universal Dependencies repositories: UD Czech-PDT (`https://github.com/UniversalDependencies/UD_Czech-PDT`) and UD English-EWT (`https://github.com/UniversalDependencies/UD_English-EWT`).

I did not train on the full treebanks. Instead, I used smaller, more manageable subsets of 2000 training sentences, 500 development sentences, and 500 test sentences for each language. This is small enough to implement, debug, and analyze carefully while still preserving a genuine train/dev/test split. Model selection still uses development data and final numbers are still reported on held-out test data.

The subset statistics confirm that the two samples differ in useful ways. The Czech subset contains 30,891 training tokens, 6,552 development tokens, and 6,862 test tokens, with average sentence lengths of 15.45, 13.10, and 13.72. The English subset contains 39,802 training tokens, 7,621 development tokens, and 7,275 test tokens, with average sentence lengths of 19.90, 15.24, and 14.55. So English is somewhat longer on average, while Czech remains the language with richer overt morphology.

The three representation settings are defined as follows. The lexicalized parser uses only word identity. The morphosyntax-only parser uses only gold UPOS and gold UFeats. The mixed parser uses lexical and morphosyntactic information together. Gold UPOS and UFeats are used deliberately so that the parsing question is isolated from tagging noise. That makes it easier to interpret the final comparison as a difference in available syntactic signal rather than as a side effect of upstream prediction errors.

3 Method

The parser is a simple graph-based dependency parser implemented directly in PyTorch. Each sentence is converted into aligned parser inputs consisting of token text, word ids, UPOS ids, feature ids, gold head indices, and gold dependency labels. A synthetic ROOT token is inserted at the beginning of each sentence so that the model always has a fixed top node for the dependency tree.

Vocabularies are built for words, part-of-speech tags, features, and labels from the training data. Morphological feature bundles are converted into stable string representations so they can be treated as discrete symbols. Sentences are padded into mini-batches, and a mask marks which positions are real tokens and which are padding. The same mask is later reused to block illegal arcs during training and decoding.

The vocabularies were built jointly from the Czech and English training subsets, so the two languages share one word vocabulary rather than using separate language-specific lexicons. This choice keeps the implementation simple and does not affect the morphosyntax-only setting, but it may slightly disadvantage the lexicalized and mixed settings compared with language-specific vocabularies.

The model has two main scoring stages. First, a token encoder produces one vector per token. In the lexicalized setting, this vector comes from word embeddings only. In the morphosyntax-only setting, it comes from UPOS and feature embeddings only. In the mixed setting, the three sources are combined. Second, an arc scorer assigns a score to every possible head-dependent pair in the sentence. A separate label scorer predicts the dependency relation once a head has been chosen.

Training uses two losses. The arc loss teaches the model to predict the correct head among all legal positions, while the label loss teaches it to predict the correct dependency label. These losses are summed into the total training objective. The training loop is fully explicit so that the parser pipeline remains visible instead of being hidden inside a higher-level trainer.

Small sanity checks are included before the final experiments to help demonstrate that the system was built and understood rather than only assembled from ready-made parser code.

Greedy decoding is used as a baseline. It predicts the highest-scoring head for each token independently and then assigns labels. This is useful as a first end-to-end check, but it does not guarantee a single well-formed dependency tree. The final parser therefore uses MST decoding. Arc scores are converted into a directed graph, and maximum spanning tree decoding selects the highest-scoring legal tree with a fixed ROOT.

All components from data reading and batching through the parser modules, training loop, decoding, and evaluation were implemented directly. Only `conllu` and `networkx` were used as utilities for CoNLL-U reading and MST graph routines.

4 Evaluation Design

The evaluation protocol is the same for every language and every representation setting. The parser is trained on the 2000-sentence training subset, evaluated after each epoch on the 500-sentence development subset, and the best epoch is selected by development LAS. Only after that choice is fixed are the final UAS and LAS values reported on the 500-sentence held-out test subset. This separation matters because it keeps the final reported numbers from being tuned directly on the test set.

Punctuation is excluded from the final reported UAS and LAS, matching standard UD evaluation practice. Also a greedy baseline is computed before switching to MST decoding. On the Czech development subset, greedy decoding without punctuation reached UAS 0.4692 and LAS 0.4252, while MST decoding improved this to UAS 0.4768 and LAS 0.4327. The gain is small but consistent, which supports using tree-constrained decoding for the main experiments.

UAS and LAS are the main metrics, but they are not the only evaluation. A single score table cannot show which labels are confused or whether improvements hold only on short local dependencies. For that reason, label-confusion analysis and dependency-length analysis are included. These extra views are still lightweight, but they make the analysis more informative and help explain why the main results look the way they do.

5 Results

5.1 Main comparison

Table 1 gives the held-out test results after development-set model selection.

Language	Representation	UAS	LAS
Czech	lexicalized	0.2189	0.1318
Czech	morphosyntax-only	0.4382	0.4017
Czech	mixed	0.4212	0.3749
English	lexicalized	0.2542	0.1770
English	morphosyntax-only	0.3409	0.2871
English	mixed	0.3398	0.2949

Table 1: Held-out test results without punctuation.

The main pattern is clear: lexicalized is the weakest setting in both languages, while morphosyntax-only and mixed are much stronger. The result is not obvious though, because the best setting differs across languages. Czech is best under morphosyntax-only input, while English is slightly best under mixed input. That difference is exactly what makes the comparison interesting.

5.2 Czech results

The Czech results most strongly support the original hypothesis. Morphosyntax-only reaches test UAS 0.4382 and LAS 0.4017, which is the best setting by a clear margin. Mixed is second with UAS 0.4212 and LAS 0.3749, while lexicalized is far behind with UAS 0.2189 and LAS 0.1318. The LAS gain of morphosyntax-only over lexicalized is 0.2699.

This suggests that overt Czech morphosyntax already contains a large amount of useful syntactic information. In this setting, the parser recovers much more structure from UPOS and UFeats than from lexical identity alone. It is also notable that mixed does not overtake morphosyntax-only in Czech, which suggests that the added lexical signal is not the main source of parsing strength in this simple architecture.

5.3 English results

The English results are weaker in absolute terms and somewhat different in ranking. Mixed is slightly best with test UAS 0.3398 and LAS 0.2949, while morphosyntax-only is very close with UAS 0.3409 and LAS 0.2871. Lexicalized remains clearly weaker at UAS 0.2542 and LAS 0.1770. So morphosyntax-only still beats lexicalized by a clear margin, but lexical identity adds a little more value in English than it did in Czech.

5.4 Cross-language comparison

The most direct cross-language comparison is the LAS gain of morphosyntax-only over lexicalized. In Czech, that gain is 0.2699. In English, it is 0.1102. Morphosyntax-only also reaches a higher absolute LAS in Czech (0.4017) than in English (0.2871). Together, these findings support the central claim that morphosyntactic information contributes much more strongly in Czech than in English under the current parser setup.

Comparison	Czech	English
LAS gain (morphosyntax-only − lexicalized)	0.2699	0.1102
Lexicalized LAS at distance 7+	0.0547	0.0856
Morphosyntax-only LAS at distance 7+	0.2278	0.1065

Table 2: Compact cross-language summary of the main difference.

At the same time, the result only partly confirms the strongest possible version of the hypothesis. If morphosyntax were equally dominant in both languages, we would expect morphosyntax-only to win clearly in English as well. Instead, English mixed is slightly best. So the final conclusion is that lexical identity still matters, but its relative importance is smaller in Czech.

5.5 Label confusions

The label-confusion analysis helps explain the score differences. In Czech, the lexicalized model often collapses very different labels into `amod`. For example, `nmod` → `amod` occurs 59 times and `conj` → `amod` occurs 35 times. There are also cases such as `obj` → `amod`. This looks like a weak and overly generic labeling strategy.

The Czech morphosyntax-only model still makes errors, but they are more structured. Common confusions include `obl` → `obj` (23 cases), `obl:arg` → `obl` (18), and `obl` → `obl:arg` (12). These are narrower confusions among related relations rather than a broad collapse into one default label.

English again shows a different pattern. The English morphosyntax-only model still often confuses `obl`, `obj`, and `nsubj`, including `obl` → `obj` (73), `nsubj` → `obj` (61), and `obj` → `nsubj` (13). Overall, the error analysis supports the broader result: Czech morphosyntax-only errors are narrower and more local, while English still lacks enough overt marking to separate some central grammatical functions as cleanly.

5.6 Dependency length

Dependency-length analysis gives another view of the same pattern. In both languages, short dependencies are easier than long ones. For example, Czech lexicalized gets LAS 0.1788 on distance-1 dependencies but only 0.0547 on dependencies of length 7+, while English lexicalized drops from 0.2040 at distance 1 to 0.0856 at distance 7+.

The important result is that morphosyntax-only beats lexicalized in every distance bin. In Czech, the difference stays large across the entire range. At distance 1, morphosyntax-only reaches LAS 0.5118 versus 0.1788 for lexicalized. At distance 7+, it still reaches 0.2278 versus 0.0547. So the gain is not limited to easy local attachments.

English shows the same direction, but a much smaller long-distance margin. At distance 1, English morphosyntax-only reaches LAS 0.3817 compared with 0.2040 for lexicalized. At distance 7+, the gap shrinks to 0.1065 versus 0.0856. This again supports the main interpretation: morphosyntax helps in both languages, but Czech retains a much clearer advantage on longer dependencies.

6 Discussion and Limitations

Taken together, the experiments support the main claim: dependency structure can be recovered to a substantial degree from morphosyntactic information alone, and that possibility is stronger in Czech than in English. The same broad pattern appears in the main UAS/LAS results, the label-confusion analysis, and the dependency-length analysis, which makes the Czech effect more convincing than a single score difference would.

At the same time, the results do not show that lexical information is unimportant or that morphosyntax-only parsing is universally best. English mixed input is slightly better than English morphosyntax-only input, so lexical identity still contributes useful information there. For that reason, the hypothesis is only partly confirmed. The main conclusion is that morphosyntax is much more informative in Czech. Lexical cues still matter in both languages.

Several limitations should be noted. The experiment is intentionally small-scale and uses 2000/500/500 subsets rather than full training sets, so the final scores are based on held-out subsets rather than full-dataset training. The parser architecture is also deliberately simple, and all comparison models were trained for a fixed 8 epochs without a patience-based early-stopping scheme. In both languages, the lexicalized model was still improving on development LAS at the end of training, so its reported performance is likely conservative. This means that the advantage of morphosyntax-only input may be somewhat overstated under the current setup. In addition, there was no broad hyperparameter search. Even so, the result remains non-trivial because the same cross-language pattern appears across several analyses rather than in only one metric.

7 Conclusion and Follow-up

This project asked whether dependency structure can be recovered from morphosyntactic information alone, and whether that ability differs between Czech and English. The answer is yes to both, with a qualification. Morphosyntax-only input strongly outperforms lexicalized input in both languages, which shows that overt grammatical information carries substantial syntactic signal. But the gain is much larger in Czech than in English, and English mixed input is still slightly best overall. So the results support the view that overt morphology is a stronger standalone cue in Czech than in English.

If more time were available, the most useful next steps would be to train on larger subsets or full training data, test a stronger contextual encoder, and run targeted feature ablations. Czech case-related morphology would be a particularly natural target, because case is a plausible source of the stronger morphosyntax-only advantage observed here. These follow-up steps would deepen the analysis without changing the core contribution of the current project.